pfi

Python for Fixed Income

COURSE 4 · MACHINE LEARNING

Scaling and K-Means

WEEK 3 · DAY 2

Created by **Jeff Lenamon**

This course is educational content created in a personal capacity. Nothing here is investment advice or a recommendation to buy or sell any security. All portfolio data is synthetic.

From a target to no target

Days 1 to 11: a target

A spread, a default flag, a yield. A function fitted to the target, judged by its errors on rows it never saw; on dated rows, later rows.

Today: no target

Groups of similar bonds across four columns at once: log OAS, duration, coupon, rating score. 821 bonds, 150 invented issuers.

Synthetic data: a cluster here is a group of similar rows in an invented file, and nothing more.

Learning without a target

	Days 1 to 11	Today
Fitted to	A target column	The columns themselves
Returns	A fitted value or probability	A cluster number per bond
Judged by	Errors on test rows	Tightness, separation, judgement
Rows put away	Test rows, first	None: nothing held out

Scaling on all bonds is not leakage today: there are no test rows. As a model feature, it would go inside the pipeline.

Distance, and why units decide it

```
distance = sqrt( (difference in col 1)^2 + (difference in col 2)^2 + ... )
```

In the file's units: distance 39.02

Column	Difference	Share (%)
oas	39.000	99.90
duration	0.834	0.05
coupon	0.875	0.05
score	0.000	0.00

In standard deviations: distance 0.6244

Column	Difference (sd)	Share (%)
oas	0.210	11.35
duration	0.257	16.88
coupon	0.529	71.77
score	0.000	0.00

1 bp counts as much as 1 year: unscaled, the biggest unit decides the distance.

K-means, step by step

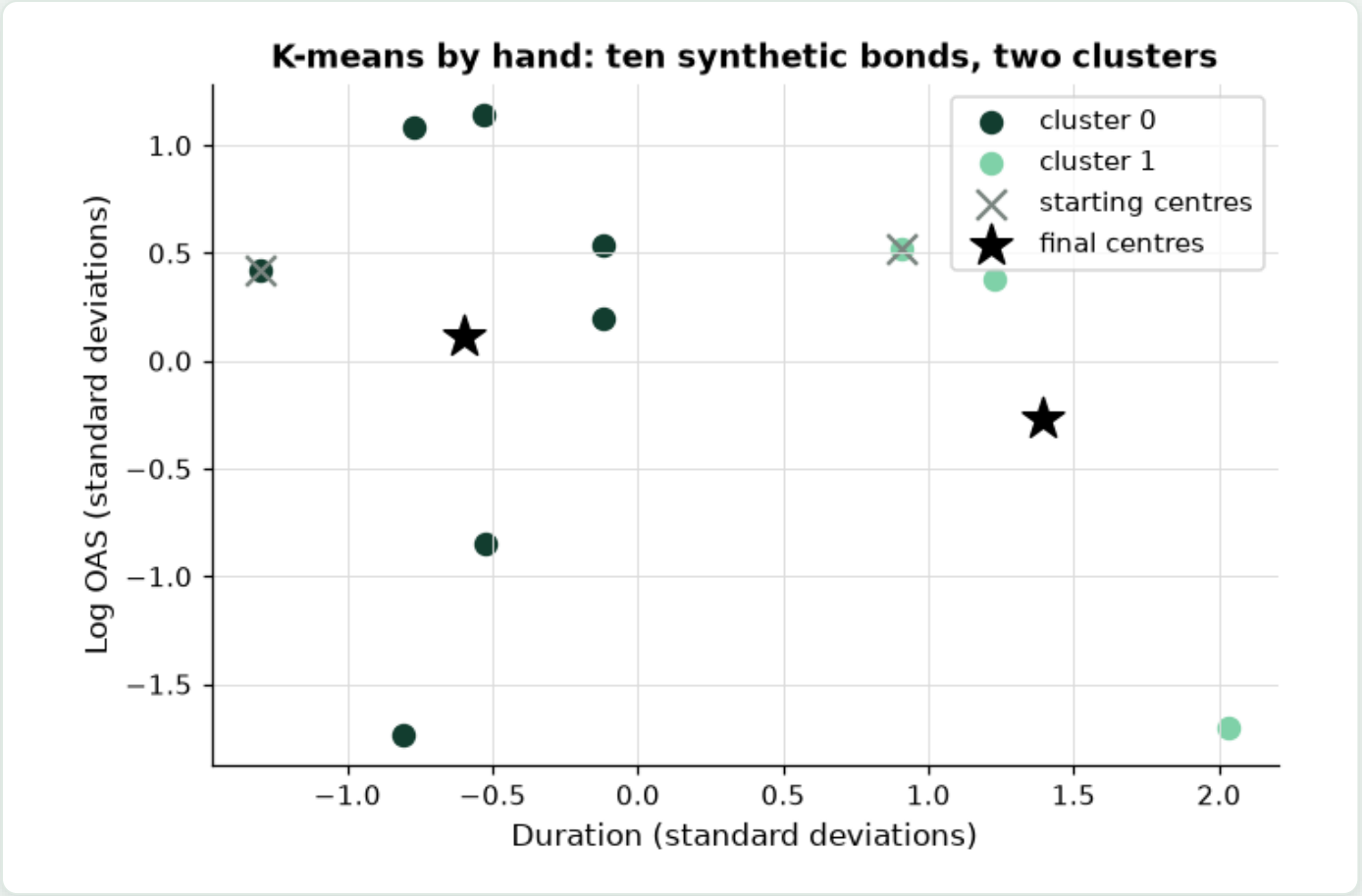
1. Start — k and k starting centres

2. Assign — each bond to nearest centre

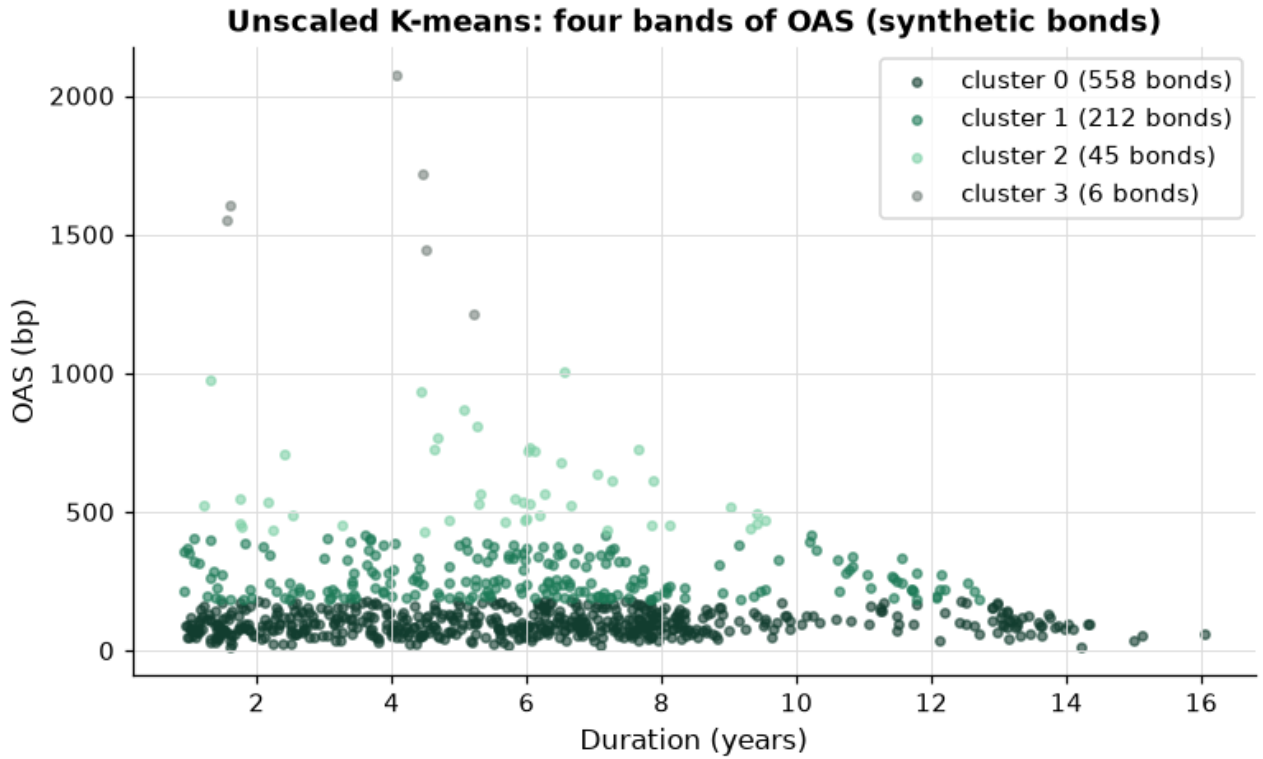
3. Update — centre = its bonds' average

4. Repeat — until no centre moves

Round	Sizes	Inertia
1	5 and 5	13.812
2	7 and 3	11.402
3	7 and 3	11.402



Unscaled: four bands of OAS



How clusters are numbered

By size, largest first: cluster 0 is the largest. A number names a group, not a score.

Cluster	Bonds	OAS (bp)	Duration
0	558	15 to 181	0.9 to 16.0
1	212	182 to 418	0.9 to 12.7
2	45	427 to 1004	1.2 to 9.5
3	6	1215 to 2074	1.5 to 5.2

Cut by OAS alone: not duration groups, not rating groups.

Before and after scaling

10.22

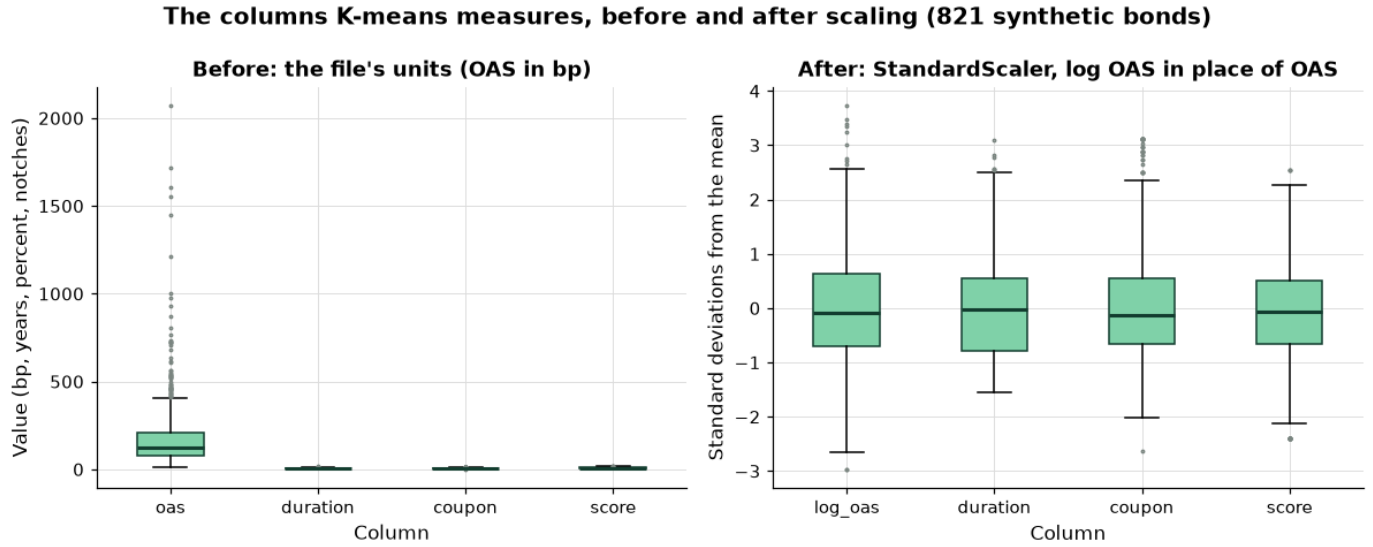
sd above the mean: widest bond, OAS in bp

3.74

sd above the mean: in logs

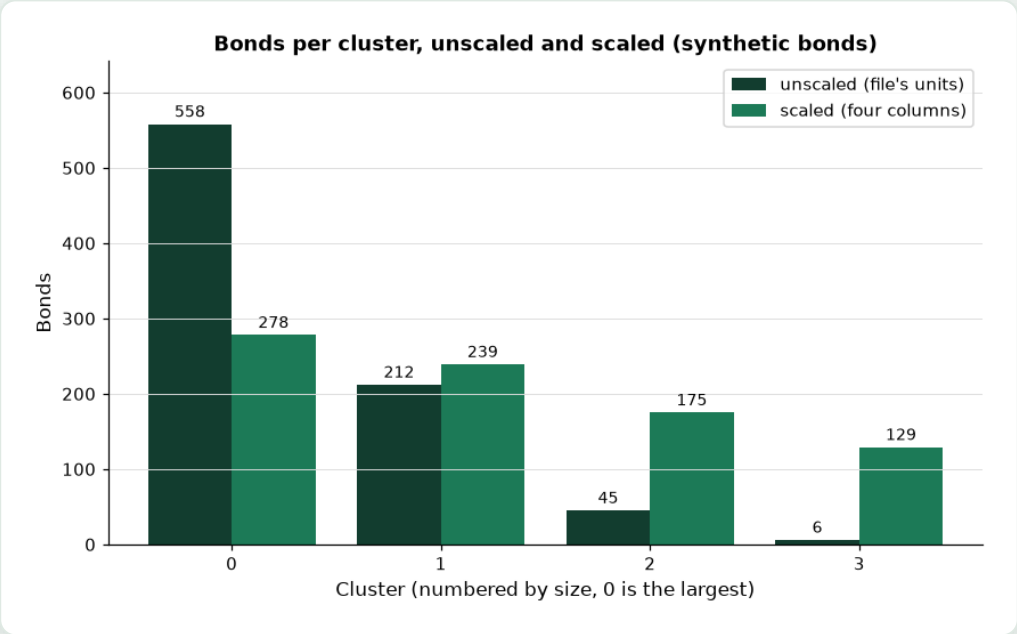
One axis, two scales

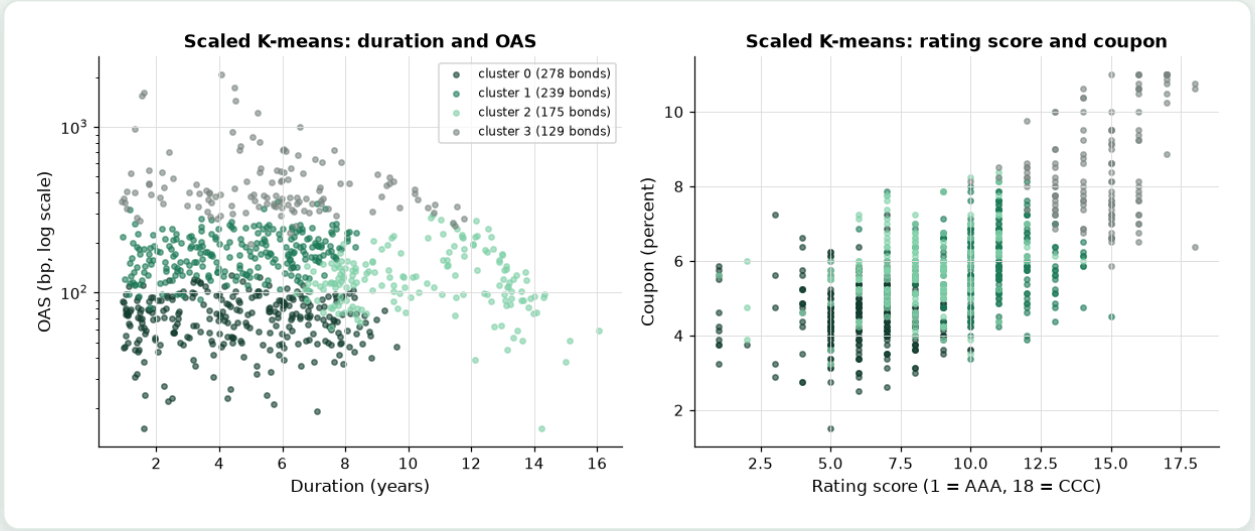
Before: OAS's middle half 79 to 210 bp, widest 2,074 bp; the other three columns stay at or below 18. After: four boxes around 0.



Scaling fixes the units, not the shape: the clusters use log OAS, as days 4 and 5 did.

Scaled, then K-means





Sizes 278, 239, 175, 129, inertia 1141.83. The ranges overlap in every column: each cluster is a region of four columns.

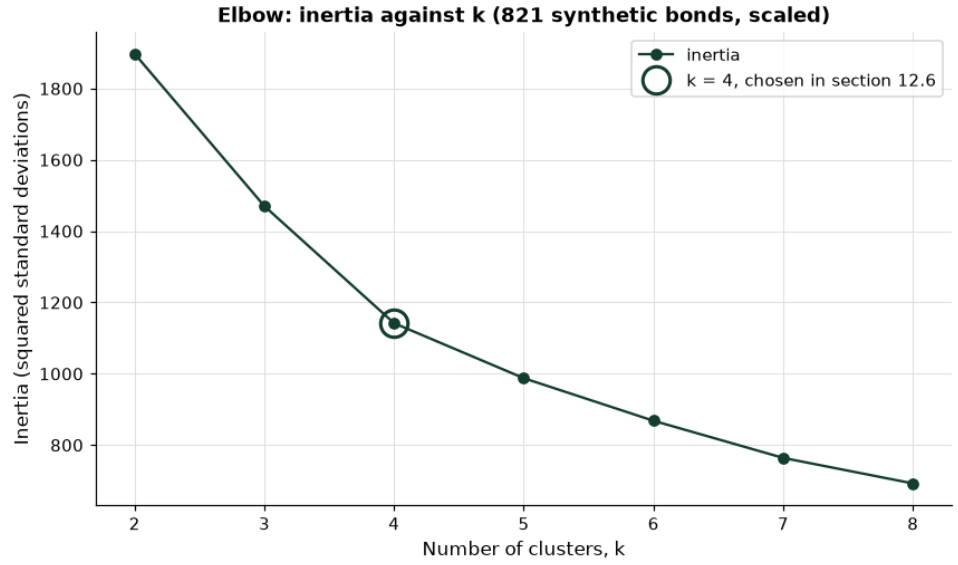
The elbow and the silhouette

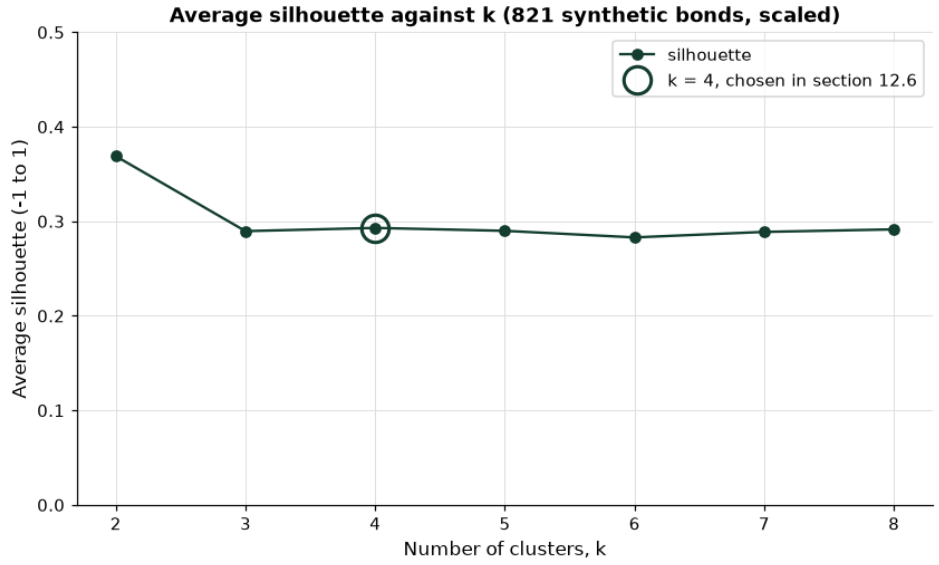
Inertia: always falls as k rises

The elbow is where one more cluster stops buying much.

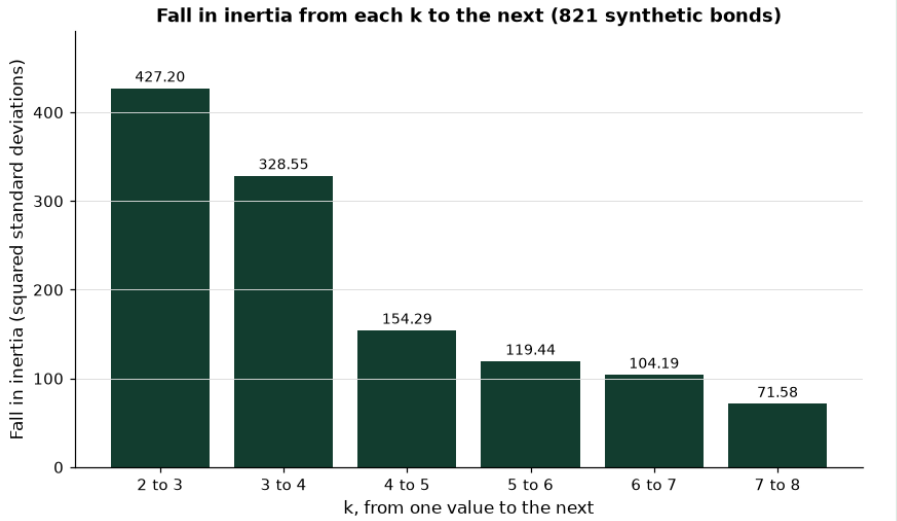
Silhouette = (b - a) / max(a, b)

a: own cluster; b: nearest other. First bond: a 1.4057, b 1.6378, silhouette 0.1417. All bonds: 0.2930. elbow_table and three tests: 50 passed.





Choosing k as a judgement



328.55 into k = 4, then 154.29 into k = 5: the fall halves.

What k = 2 separates

538 and 283 bonds: OAS 96.22 against 336.60 bp, score 7.34 against 12.88, duration 6.00 against 5.95 years. Duration plays no part; k = 4 adds a duration group.

The course takes k = 4

The curve bends there and its silhouette is the highest above 2. k = 2 or 5 could be argued as well.

The charts support a choice. Neither discovers a true number of clusters.

Clusters described, never ranked

Allowed names

cluster, cluster 2

a group of similar rows in an invented dataset

described by counts and averages

Never called

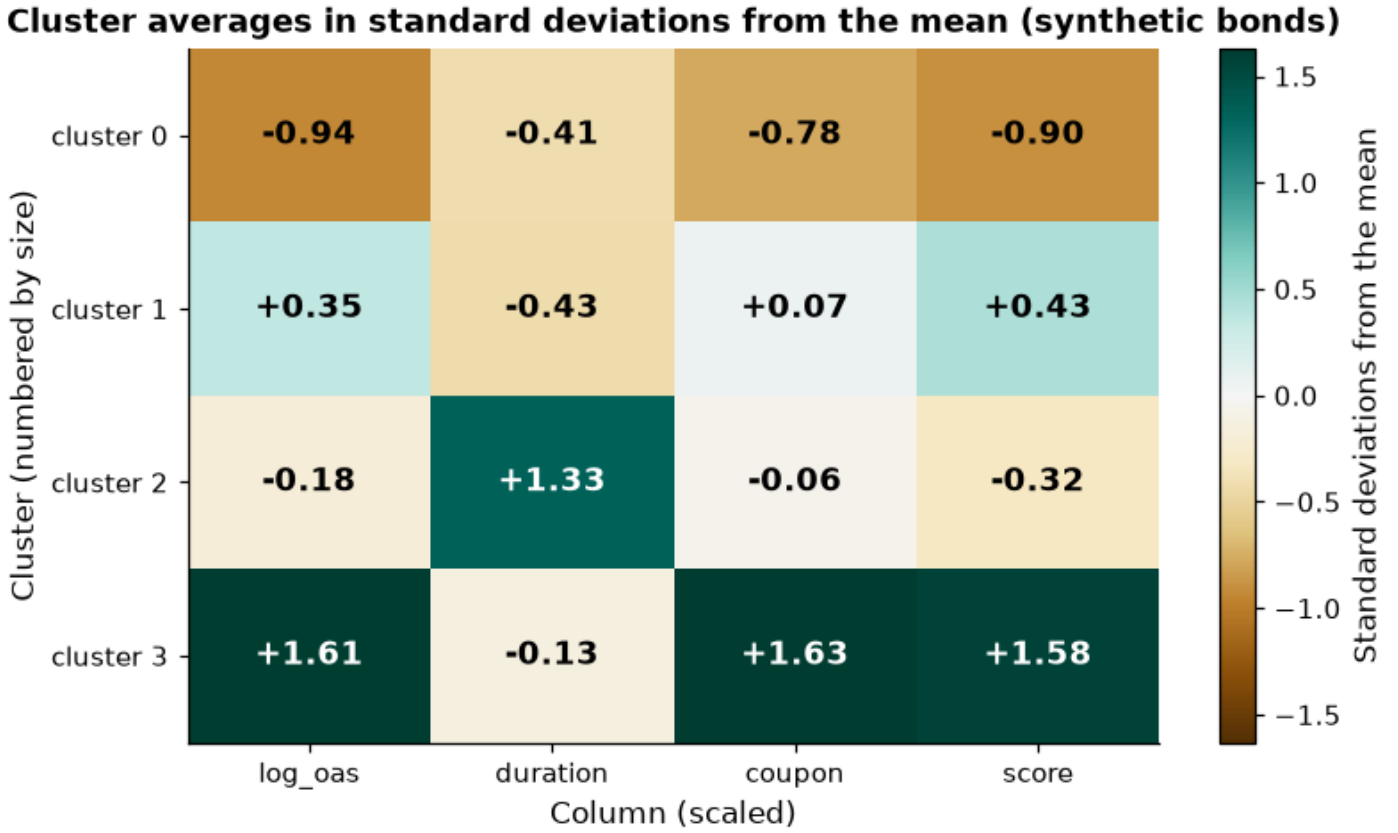
a recommendation, a ranking, a quality label

good, bad, safe, risky

cheap, rich; a peer group of real issuers

Cluster	Bonds	OAS (bp)	Duration (yrs)	Coupon (%)	Score	IG (%)
0	278	70.8	4.65	4.57	6.16	100.0
1	239	178.5	4.59	5.96	10.74	45.6
2	175	126.3	10.32	5.75	8.14	86.9
3	129	485.0	5.55	8.55	14.66	0.8

Cluster averages on one scale



Read across a row

Cluster 2: +1.33 on duration, near 0 elsewhere. Cluster 3: +1.58 to +1.63 on log OAS, coupon, and score. Cluster 0: below the average bond on all four.

Distance from the average bond, either way. A description, not a ranking.

Excel side by side

Excel side by side (tested in Excel)

`=STANDARDIZE(A2,AVERAGE(A$2:A$822),STDEV.P(A$2:A$822))`	StandardScaler, within 1e-9
`=STANDARDIZE(..., STDEV.S(...))`	smaller by 1.00061: not a match
`=SQRT(SUMXMY2($G2:$J2,$T$1:$W$1))`	distance to centre 0
`=MATCH(MIN(L2:O2),L2:O2,0)-1`	nearest centre: 821 of 821
`=AVERAGEIF($P$2:$P$822,$S7,G$2:G$822)`	update step: the averages
`=COUNTIF($P$2:$P$822,$S7)`	sizes 278, 239, 175, 129

Excel has no K-means, silhouette, or elbow function. It can check every step of one round.

Save the day: git show HEAD:results.csv

git show HEAD:results.csv (the last commit)

```
model,features,k,rows,inertia,silhouette
kmeans_scaled,log_oas duration coupon score,4,821,1141.8300,0.2930
```

The working file, after adding k = 2

```
model,features,k,rows,inertia,silhouette
kmeans_scaled,log_oas duration coupon score,4,821,1141.8300,0.2930
kmeans_scaled,log_oas duration coupon score,2,821,1897.5800,0.3689
```

HEAD:file is that file inside the last commit: what the log said before the change.

The AI check

The docstring is the prompt. The code that comes back runs, and has one modelling bug.

1

Docstring

Which rows fit, which rows report, and the units: what you asked for.

→

2

Code back

Written for the course in the style of an AI answer. Read it.

→

3

Test first

Assert what the docstring promises. Run it: it fails.

→

4

Fix

Find the rule it broke, fix it, run the test again.

Today: K-means on raw OAS in bp and duration in years, read as "duration groups". Means [179.08, 5.98]: nothing scaled.

Code that runs is not the same as code that is right.

What you can do now

- No target: clustering ends in a judgement
- Unscaled, the biggest unit decides the distance
- K-means: start, assign, update, repeat
- Elbow and silhouette support a choice of k
- Clusters described by counts and averages, never ranked

Week 3, Day 3 Hierarchical clustering and cluster profiling: the 150 issuers